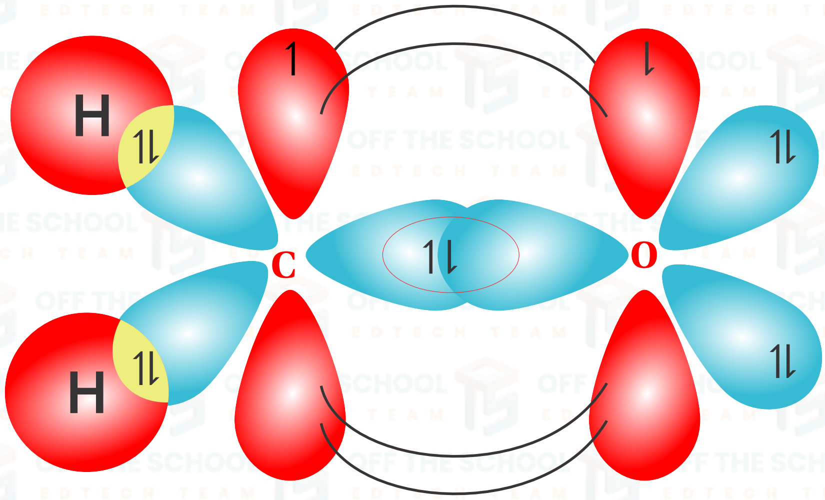
UNIT 08

CARBONYL COMPOUNDS I: ALDEHYDES & KETONES

STRUCTURE & NATURE OF CARBONYL GROUP:

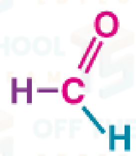
In the carbonyl group (>C=O), the carbon atom is sp^2 hybridized. One of the porbitals of the carbonyl carbon remains unhybridized. The three sp^2 hybrid orbitals of carbon lie in the trigonal plane with an angle of about 120° between them. These three sp^2 hybrid orbitals form three σ-bonds; one with the half-filled p-orbital of oxygen and the other two with the other atoms attached with this carbon. The unhybridized p-orbital of the carbonyl carbon undergoes sidewise overlapping with the remaining half-filled p-orbital of the oxygen atom to form a π-bond.

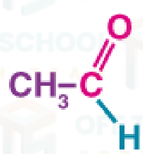


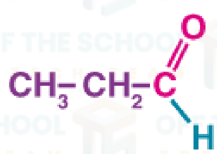
What are Aldehydes?

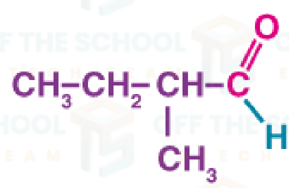
In aldehydes, the carbonyl group has one hydrogen atom attached to it together with either a 2nd hydrogen atom or a hydrogen group which may be an alkyl group or one containing a benzene ring.

Examples;


Methanal


Ethanal

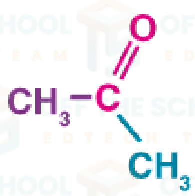

Propanal

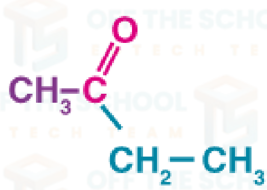

2-Methylbutanal

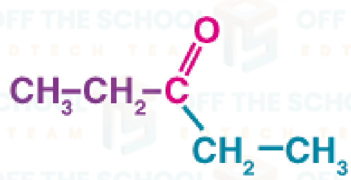
What are Ketones?

In ketones, the carbonyl group has 2 hydrocarbon groups attached to it. These can be either the ones containing benzene rings or alkyl groups. Ketone does not have a hydrogen atom attached to the carbonyl group.

Example:


Propanone

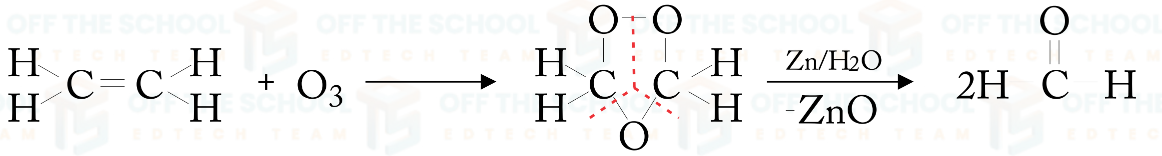

Butanine


Pentan-3-one

Preparation:

1. Ozonolysis of straight-chain alkenes:

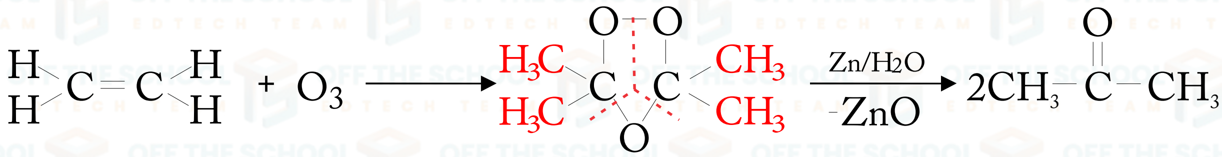
Alkene is treated with ozone in the presence of CCl_4 give an ozonide which is decomposed by water and zinc dust to give aldehyde or ketone or both. Ozonolysis of straight-chain alkenes gives aldehydes.



(ethyne) (ozonide) formaldehyde

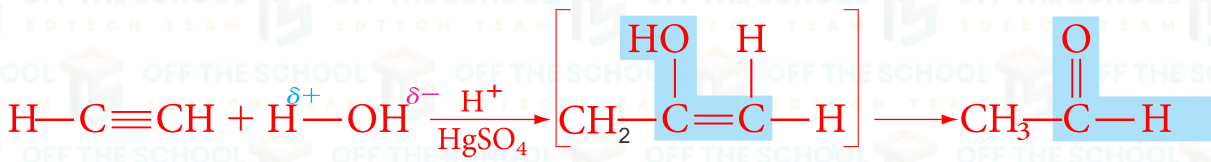
2. Ozonolysis of branched-chain alkenes:

Ozonolysis of branched-chain alkenes gives ketone.



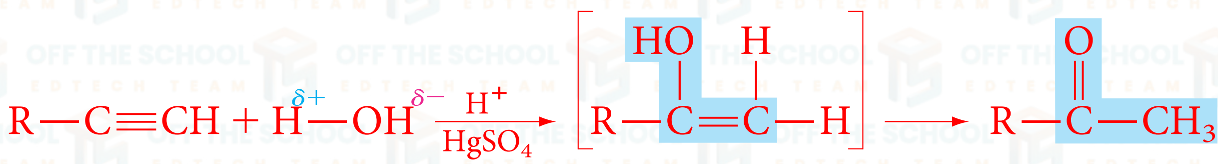
(ethyne) (ozonide) Propanone

3. Catalytic Hydration of alkyne:



a vinyl alcohol, or enol

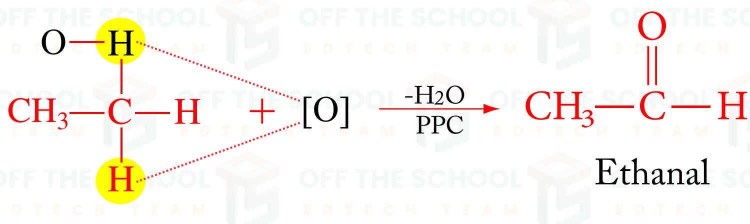
Ethanal



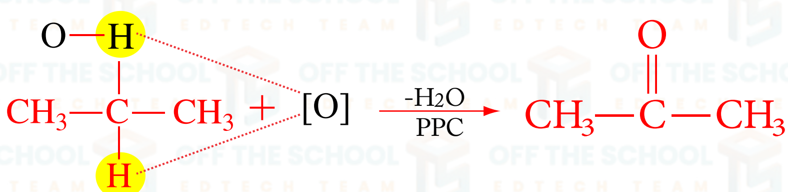
a vinyl alcohol, or enol

a methyl ketone

4. Oxidation of alcohols:



Ethanal



2-Methylpropanol

Acetone

Aromatic ketones can synthesize through Friedel–Crafts acylation of an aromatic ring with an acid chloride. Aluminum chloride ($AlCl_3$) is used as a Lewis acid catalyst.

Acetophenone

Reactivity of Carbonyl Compounds:

1. Polar Bond: The bond between carbon (C) and oxygen (O) in the carbonyl group (C=O) is polar.

2. Electronegativity: Oxygen is more electronegative than carbon, so it pulls electrons towards itself more(8).

3. Partial Charges: This makes the oxygen partially negative (δ-) and the carbon partially positive (δ+).

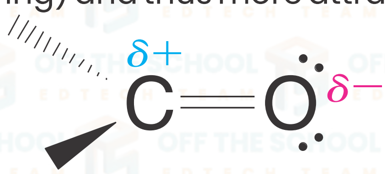
4. Reactivity: The carbon in the carbonyl group can attract nucleophiles (particles that donate electrons) because it is electron-deficient.

5. Aldehydes vs. Ketones: Aldehydes are typically more reactive than ketones.

6. Steric Hindrance: Aldehydes have less steric hindrance (less crowding around the reactive site) because they have one less alkyl group than ketones.

7. Electron Withdrawal: The hydrogen attached to the carbonyl carbon in aldehydes can pull some electron density away, making the carbon even more positive.

8. Electrophilicity: Because of this, the carbon in aldehydes is a stronger electrophile (electron-loving) and thus more attractive to nucleophiles.



Nucleophilic addition reaction:

A nucleophilic addition reaction is a type of chemical reaction that is common with compounds that have a carbon-oxygen double bond, like aldehydes and ketones.

Here's what happens in simple terms:

1. Nucleophile: This is a particle that loves to give away its electrons. It's like a friend who's always ready to share.

2. Aldehydes and Ketones: These are chemicals that have a special bond — a carbon atom double-bonded to an oxygen atom (C=O). Aldehydes have at least one hydrogen attached to the carbon, while ketones have two carbon groups attached.

3. Attraction: The carbon in this double bond is a bit positive because the oxygen likes to keep more electrons (it's more electronegative). So, the carbon is like a magnet for particles that have extra electrons — the nucleophiles.

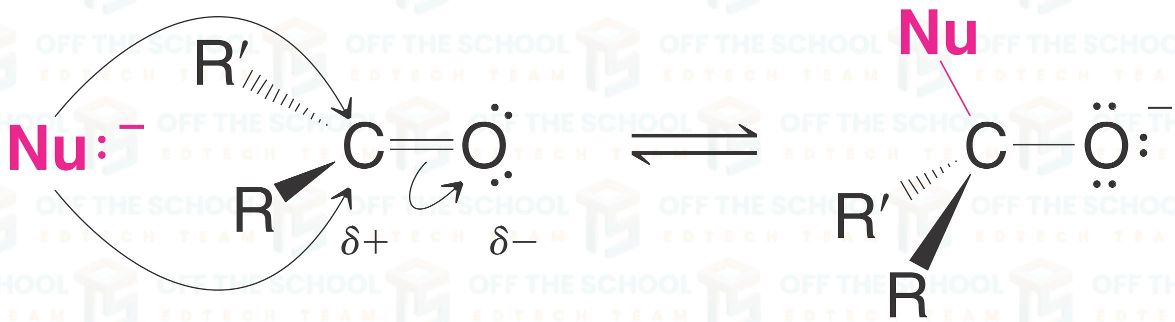
4. The Reaction: The nucleophile comes in and donates its electrons to the carbon, breaking the double bond with oxygen. This turns the double bond (C=O) into a single bond (C-O), adding the nucleophile to where the double bond used to be.

5. Result: You get a new compound where the nucleophile has joined the party, attached to the carbon, and the oxygen now has a single bond with carbon and also has an extra pair of electrons, making it negative (or part of a new bond if a proton is available to attach to it).

For a 12th-grade student, remember:

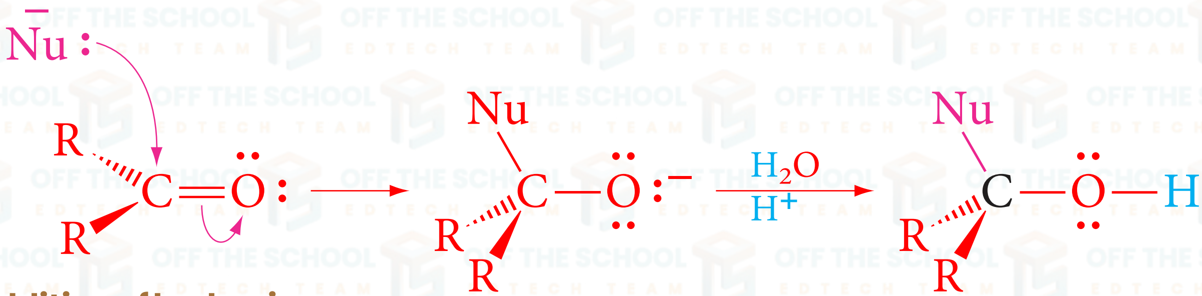
- **Nucleophile:** Gives electrons.
- **Carbon in C=O:** Positive and ready to accept electrons.
- **Addition:** Nucleophile adds to carbon, double bond becomes single bond.

There are two types of N.A.R i.e., Acid & Base-Catalyzed reactions.

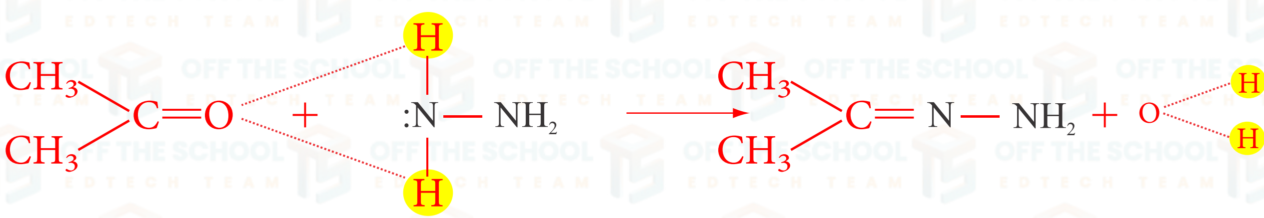


Acid-catalyzed reaction:

A acid-catalyzed reaction is a chemical reaction where a acid speeds up the rate of the reaction. The acid is not consumed in the process; it helps the reaction go faster.

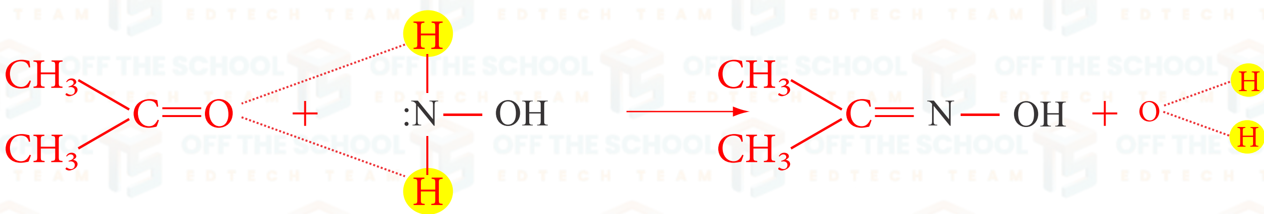


Addition of hydrazine:



Propanone Hydrazine (Acetone hydrazone) Propanone hydrazone

Addition of hydroxyl amine:



Propanone Hydroxyl amine Propanone oxime

Base catalyzed reaction:

A base-catalyzed reaction is a chemical reaction where a base speeds up the rate of the reaction. The base is not consumed in the process; it helps the reaction go faster.

Base Catalyzed Reaction Mechanism:

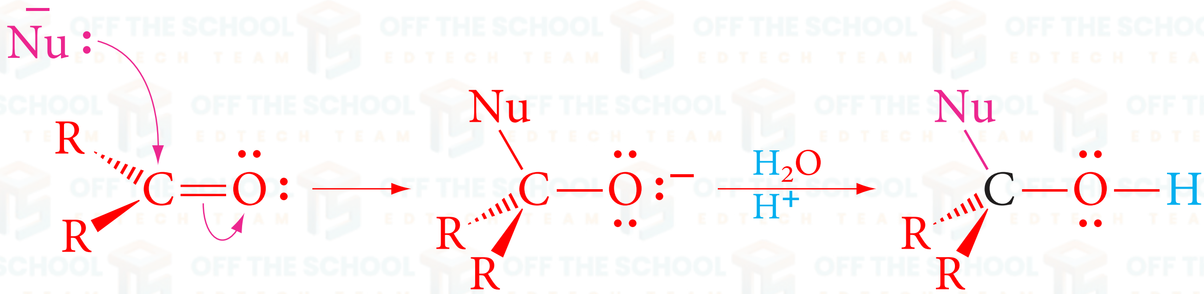
Let's explain the reaction mechanism shown in the image step by step.

Step 1: Nucleophilic Attack

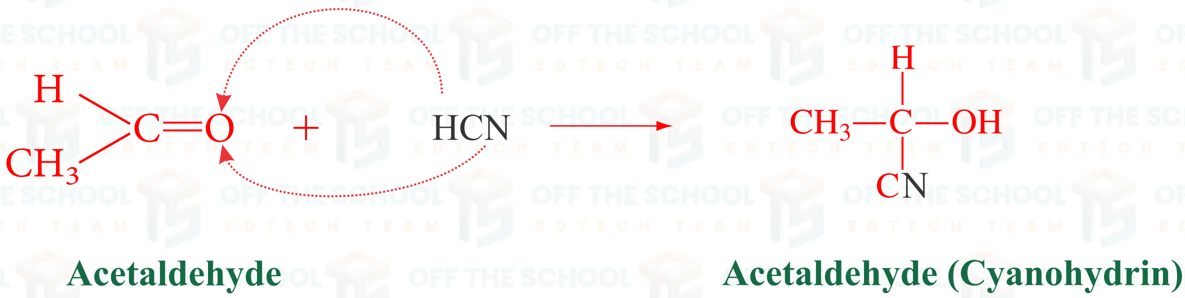
A nucleophile (Nu^-) attacks a carbonyl compound (C=O) because the carbon is slightly positive due to the oxygen pulling electron density away from the bond. The nucleophile breaks the double bond, and the oxygen gains a negative charge.

Step 2: Protonation

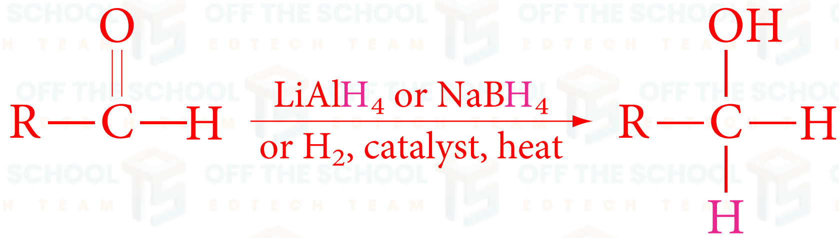
An intermediate compound forms with the oxygen negatively charged. Water (H_2O) donates a proton (H^+) to neutralize the charge on oxygen, resulting in the final product where the nucleophile is bonded to carbon, and oxygen has an OH group.



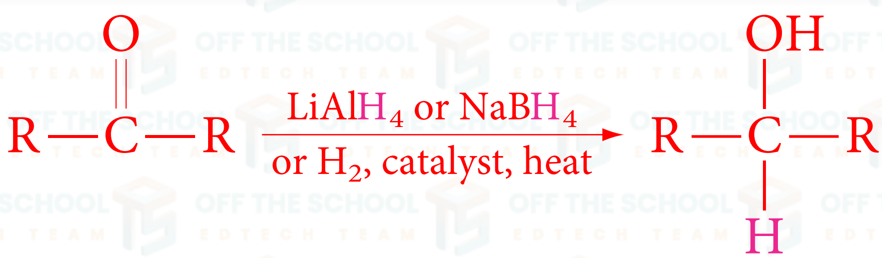
Addition of HCN:



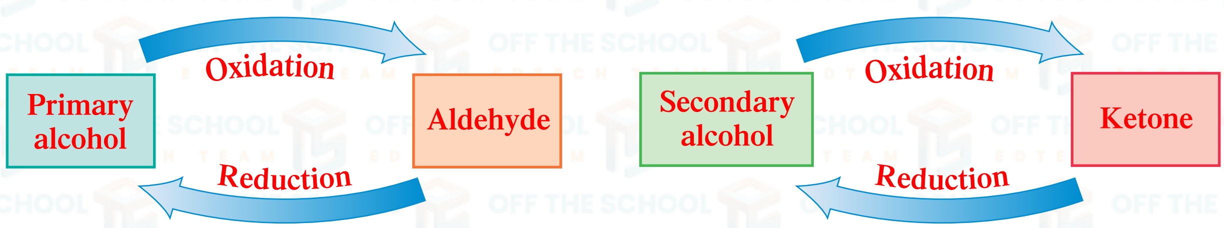
Reduction to Alcohols



1°alcohol



2°alcohol



REDUCTION TO HYDROCARBON

$$R-\overset{\overset{\displaystyle O}{\|}}{C}-H \;+\; 4\,[H] \xrightarrow[\text{HClconc.}]{\text{Zn(Hg)}} R-\underset{\underset{\displaystyle H}{|}}{\overset{\overset{\displaystyle H}{|}}{C}}-H \;+\; H_2O$$

$$R-\overset{\overset{\displaystyle O}{\|}}{C}-R \;+\; 4\,[H] \xrightarrow[\text{HCl}_{\text{conc.}}]{\text{Zn(Hg)}} R-\underset{\underset{\displaystyle H}{|}}{\overset{\overset{\displaystyle H}{|}}{C}}-R \;+\; H_2O$$

OXIDATION TO CARBOXYLIC ACIDS

$$\underset{\text{aldehyde}}{R-\overset{\overset{\displaystyle O}{\|}}{C}-H} \xrightarrow[\text{Chromicacid}]{H_2CrO_4} \underset{\text{carboxylic acid}}{R-\overset{\overset{\displaystyle O}{\|}}{C}-OH}$$

$$\underset{\text{aldehyde}}{R-\overset{\overset{\displaystyle O}{\|}}{C}-R} + 4\,[O] \xrightarrow[O_2,\ \text{or}\ Ag^{2+},\ NaOH]{CrO_3,\ H_2SO_4,\ H_2O \ \text{or}} \underset{\text{carboxylic acid}}{R-\overset{\overset{\displaystyle O}{\|}}{C}-OH}$$

Distinguish test between Aldehydes and Ketones:

Silver Mirror Test (Tollen's Test)

1. Reagent Preparation: Tollen's reagent is prepared by mixing silver nitrate with ammonia, which produces a complex ammoniacal silver nitrate solution.

2. Reaction with Aldehyde: When an aldehyde is warmed with Tollen's reagent, it gets oxidized, and the solution reduces silver ions to metallic silver.

3. Observation: If an aldehyde is present, you'll see a shiny, mirror-like coating of silver on the inside of the test tube.

$$\underset{\text{Ethanal (acetaldehyde)}}{CH_3-\overset{\overset{\large O}{\|}}{C}-H} + \underset{\text{Tollens' reagent}}{2Ag^+} \xrightarrow{[O]} \underset{\text{Silver mirror}}{2Ag(s)} + \underset{\text{Ethanoic acid (acetic acid)}}{CH_3-\overset{\overset{\large O}{\|}}{C}-OH}$$

Fehling's Test:

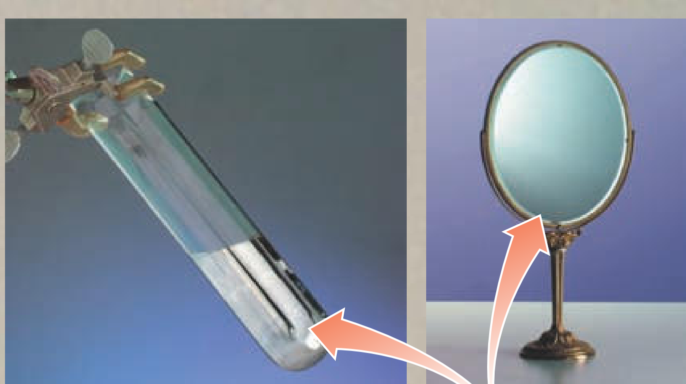
1. Solution Preparation: Fehling's solution is made up of two parts: Fehling A, which is a copper(II) sulphate solution, and Fehling B, which is an alkaline sodium potassium tartrate solution in sodium hydroxide.

2. Mixing: Equal volumes of Fehling A and B are mixed in a test tube.

3. Reaction with Aldehyde: The mixed solution is heated and then a sample of the substance being tested is added. If an aldehyde is present, the blue copper (II) ions in the solution are reduced to red copper(I) oxide.

Observation: The presence of aldehyde is indicated by a change from the blue colour of the solution to a red precipitate of copper(I) oxide.

$$\underset{\text{2-Hydroxypropanal}}{CH_3-\overset{\overset{\large OH}{|}}{CH}-\overset{\overset{\large O}{\|}}{C}-H} + \underset{\text{Benedict's reagent}}{2Cu^{2+}} \longrightarrow \underset{\text{Brick-red solid}}{Cu_2O(s)} + \underset{\text{2-Hydroxypropanoic acid}}{CH_3-\overset{\overset{\large OH}{|}}{CH}-\overset{\overset{\large O}{\|}}{C}-OH}$$



$$Ag^+ + 1\,e^- \longrightarrow Ag(s)$$

In Tollens' test, a silver mirror forms when the oxidation of an aldehyde reduces silver ion to metallic silver. The silvery surface of a mirror is formed in a similar way.

Cu^{2+} $Cu_2O(s)$

The blue Cu^{2+} in Benedict's solution forms a brick-red solid of Cu_2O in a positive test for many sugars and aldehydes.